

```

#include <Arduino_GFX_Library.h>

#include <Usg2lib.h>
#define TFT_BL 9
Arduino_DataBus *bus = new Arduino_ESP8266SPI(12 /* DC */, 2 /* CS */);
//Arduino_DataBus *bus = new Arduino_ESP8266SPI(4 /* DC */, 2 /* CS */, 18 /* SCK */, 23 /* MOSI */, -1 /* MISO */);
Arduino_GFX *gfx = new Arduino_GC9A01(bus, 33 /* RST */, 3 /* rotation */, true /* IPS */);

#include "FreeMono9pt7b.h"
#include "FreeSansBold10pt7b.h"
#include "FreeSerifBoldItalic12pt7b.h"

```

Fig. 3: Setting the board and pins for display

```

#define NTP_DBG_PORT          Serial
// Debug Level from 0 to 4
#define NTP_LOGLEVEL_        0

#include <NTPClient_Generic.h>

#if (ESP32)
#include <WiFi.h>
#elif (ESP8266)
#include <ESP8266WiFi.h>
#endif

#include <WiFiUdp.h>

char ssid[] = "JioFiber-4G";          // your network SSID (name)
char pass[] = "1234567890";          // your network password

WiFiUDP ntpUDP;

#define TIME_ZONE_OFFSET_HRS    (-4)

NTPClient timeClient(ntpUDP;

```

Fig. 4: Wi-Fi setting for NTP server

round shaped LCD (MOD1), MAX 30100 (MOD2), ESP12F (MOD3), MCP73812T (MOD), and a few other components. See Bill of Material table for the complete list of components.

Connect the components as per circuit diagram with the battery + ve and -ve terminals connected to jumper JP1. The left side of the circuit in Fig. 2 is for battery charging. To charge the battery, move slide

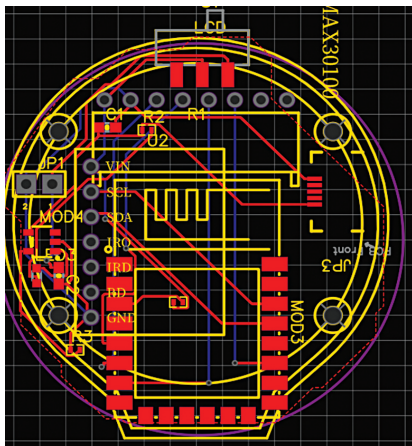


Fig. 5: Watch PCB design

switch S1 connected to JP1 pin to its off side and charge the battery using a micro USB charger. After charging, move the switch back to its on position.

Software

Prepare code for the smart watch so that it can accept the data and show it the way you want. To use Arduino for coding,

download the latest Arduino IDE. Since the IDE requires an ESP board to program, go to ESP website and follow the instructions there to install the ESP board on Arduino IDE.

We need various libraries to interface with the LCD and for getting time from the network. So, open Library Manager and install NTP client, MAX30100, and Arduino GFX libraries.

After installing libraries, interface the round LCD display using Arduino GFX library and include that in the code. We need the NTP client to get time from the Wi-Fi network, so include that as well in the code. As we need to interface the MAX30100 heart sensor, so include that library as well.

After including all the libraries, set up the code for the hardware SPI bus pins and board type. Here we are using the ESP826612F chip, so set the board as ESP8266 and then set the DC and CS pins of the LCD display. You can use any free pin on ESP CH1.

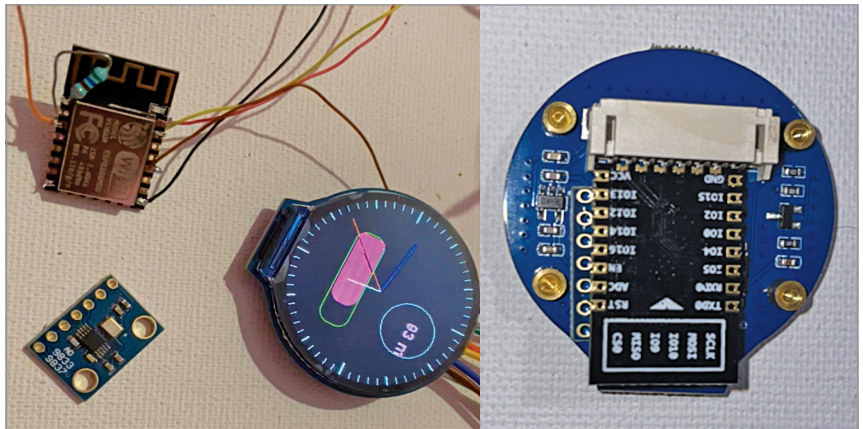


Fig. 6: Soldering ESP12F chip with display and placing the ESP chip behind watch

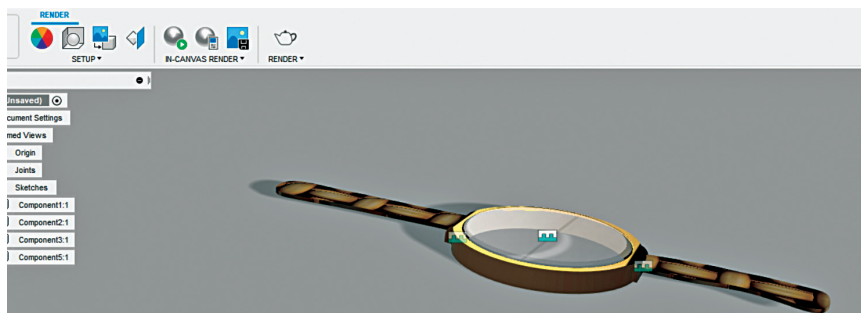


Fig. 7: Smart watch 3D case design